

Администрирование локальных сетей

Лабораторная работа №12. Настройка NAT

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Цели и задачи работы

Приобрести практические навыки по настройке доступа локальной сети к внешней сети посредством NAT.

- Выполнить первоначальную настройку оборудования провайдера.
- Настроить интерфейсы маршрутизатора и коммутатора провайдера.
- Настроить подключение сети «Донская» к сети провайдера.
- Настроить NAT и правила доступа.
- Проверить доступность внешних ресурсов и внутренних серверов.

Выполнение лабораторной работы

Настройка маршрутизатора провайдера

```
provider-babdullakhi-gw-1(config)#line console 0
provider-babdullakhi-gw-1(config-line)#pass cisco
provider-babdullakhi-gw-1(config-line)#login
provider-babdullakhi-gw-1(config-line)#exit
provider-babdullakhi-gw-1(config)#enab secret cisco
provider-babdullakhi-gw-1(config)#serv
provider-babdullakhi-gw-1(config)#service pas
provider-babdullakhi-gw-1(config)#service password-encryption
provider-babdullakhi-gw-1(config)#user admin priv 1 secret cisco
provider-babdullakhi-gw-1(config)#
provider-babdullakhi-gw-1(config)#int f0/0
provider-babdullakhi-gw-1(config-if)#no shut

provider-babdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

provider-babdullakhi-gw-1(config-if)#int f0/0.4
provider-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.4, changed state to up

provider-babdullakhi-gw-1(config-subif)#enc dot1q 4
provider-babdullakhi-gw-1(config-subif)#ip address 198.51.100.1 255.255.255.240
provider-babdullakhi-gw-1(config-subif)#descr msk-donskaya
provider-babdullakhi-gw-1(config-subif)#exit
provider-babdullakhi-gw-1(config)#int f0/1
provider-babdullakhi-gw-1(config-if)#no shut

provider-babdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

provider-babdullakhi-gw-1(config-if)#ip addr 192.0.2.1 255.255.255.0
provider-babdullakhi-gw-1(config-if)#descr internet
provider-babdullakhi-gw-1(config-if)#exit
provider-babdullakhi-gw-1(config)#exit
provider-babdullakhi-gw-1#
%SYS-5-CONFIG_1: Configured from console by console

provider-babdullakhi-gw-1#wr mem
Building configuration...
[OK]
provider-babdullakhi-gw-1#
```

Настройка коммутатора провайдера

```
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host provider-babdullakhi-sw-1
provider-babdullakhi-sw-1(config)#line vty 0 4
provider-babdullakhi-sw-1(config-line)#pass cisco
provider-babdullakhi-sw-1(config-line)#login
provider-babdullakhi-sw-1(config-line)#line cons 0
provider-babdullakhi-sw-1(config-line)#pass cisco
provider-babdullakhi-sw-1(config-line)#login
provider-babdullakhi-sw-1(config-line)#exit
provider-babdullakhi-sw-1(config)#enab secret cisco
provider-babdullakhi-sw-1(config)#serv
provider-babdullakhi-sw-1(config)#service pas
provider-babdullakhi-sw-1(config)#service password-encryption
provider-babdullakhi-sw-1(config)#user admin priv 1 secret cisco
provider-babdullakhi-sw-1(config)#int f0/1
provider-babdullakhi-sw-1(config-if)#sw mode trunk

provider-babdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

provider-babdullakhi-sw-1(config-if)#int f0/24
provider-babdullakhi-sw-1(config-if)#sw mode trunk

provider-babdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

provider-babdullakhi-sw-1(config-if)#exit
provider-babdullakhi-sw-1(config)#vlan 4
provider-babdullakhi-sw-1(config-vlan)#name nat
provider-babdullakhi-sw-1(config-vlan)#exit
provider-babdullakhi-sw-1(config)#int vlan 4
provider-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan4, changed state to up

provider-babdullakhi-sw-1(config-if)#no shut
provider-babdullakhi-sw-1(config-if)#exit
provider-babdullakhi-sw-1(config)#exit
provider-babdullakhi-sw-1#
%SYS-5-CONFIG I: Configured from console by console
```

Подключение сети «Донская» к провайдеру

```
msk-donskaya-babdullakhi-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-babdullakhi-gw-1(config)#int f0/1
msk-donskaya-babdullakhi-gw-1(config-if)#no shut

msk-donskaya-babdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-donskaya-babdullakhi-gw-1(config-if)#int f0/1.4
msk-donskaya-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.4, changed state to up

msk-donskaya-babdullakhi-gw-1(config-subif)#enc dot1q 4
msk-donskaya-babdullakhi-gw-1(config-subif)#ip address 198.51.100.2 255.255.255.240
msk-donskaya-babdullakhi-gw-1(config-subif)#descr internet
msk-donskaya-babdullakhi-gw-1(config-subif)#exit
msk-donskaya-babdullakhi-gw-1(config)#exit
msk-donskaya-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-babdullakhi-gw-1(config)#ip route 0.0.0.0 0.0.0.0 198.51.100.1
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#ip nat pool main-pool 198.51.100.2 198.51.100.14 netmask 255.255.255.240
msk-donskaya-babdullakhi-gw-1(config)#ip access-list extended nat-inet
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark dk
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.11 eq 80
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.12 eq 80
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark departments
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.4.0 0.0.0.255 host 192.0.2.13 eq 80
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark adm
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.5.0 0.0.0.255 host 192.0.2.14 eq 80
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark admin
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit ip host 10.128.6.200 any
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit ip host 10.128.6.222 any
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#exit
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#
```

Рис. 3: Настройка интерфейса маршрутизатора сети «Донская»

Настройка NAT-пула и ACL

Создан пул адресов `main-pool` и расширенный список доступа `nat-inet`, определяющий разрешённый трафик из локальной сети во внешнюю сеть.

```
msk-donskaya-babdullakhi-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-babdullakhi-gw-1(config)#int f0/1
msk-donskaya-babdullakhi-gw-1(config-if)#no shut

msk-donskaya-babdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-donskaya-babdullakhi-gw-1(config-if)#int f0/1.4
msk-donskaya-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.4, changed state to up

msk-donskaya-babdullakhi-gw-1(config-subif)#enc dot1q 4
msk-donskaya-babdullakhi-gw-1(config-subif)#ip address 198.51.100.2 255.255.255.240
msk-donskaya-babdullakhi-gw-1(config-subif)#descr internet
msk-donskaya-babdullakhi-gw-1(config-subif)#exit
msk-donskaya-babdullakhi-gw-1(config)#exit
msk-donskaya-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-babdullakhi-gw-1(config)#ip route 0.0.0.0 0.0.0.0 198.51.100.1
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#ip nat pool main-pool 198.51.100.2 198.51.100.14 netmask 255.255.255.240
msk-donskaya-babdullakhi-gw-1(config)#ip access-list extended nat-inet
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark dk
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.11 eq 80
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.12 eq 80
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark departaments
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.4.0 0.0.0.255 host 192.0.2.13 eq 80
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark adm
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.5.0 0.0.0.255 host 192.0.2.14 eq 80
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark admin
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit ip host 10.128.6.200 any
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit ip host 10.128.6.222 any
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#exit
```

Внутренние интерфейсы были назначены как `ip nat inside`, внешний интерфейс — как `ip nat outside`.

```
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source list nat-inet pool main-pool overload
msk-donskaya-babdullakhi-gw-1(config)#int f0/0.3
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/0.101
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/0.102
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/0.103
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/0.104
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/1.4
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat outside
msk-donskaya-babdullakhi-gw-1(config-subif)#exit
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.2 80 198.51.100.2 80
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.3 20 198.51.100.3 20
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.3 21 198.51.100.3 21
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.4 25 198.51.100.4 25
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.4 110 198.51.100.4 110
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.6.200 3389 198.51.100.10 3389
msk-donskaya-babdullakhi-gw-1(config)#exit
msk-donskaya-babdullakhi-gw-1#
%SYS-S-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-babdullakhi-gw-1#
```

Рис. 5: Настройка NAT inside/outside

Настроен проброс портов для доступа к внутренним серверам из внешней сети.

```
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source list nat-inet pool main-pool overload
msk-donskaya-babdullakhi-gw-1(config)#int f0/0.3
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/0.101
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/0.102
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/0.103
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/0.104
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat inside
msk-donskaya-babdullakhi-gw-1(config-subif)#int f0/1.4
msk-donskaya-babdullakhi-gw-1(config-subif)#ip nat outside
msk-donskaya-babdullakhi-gw-1(config-subif)#exit
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.2 80 198.51.100.2 80
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.3 20 198.51.100.3 20
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.3 21 198.51.100.3 21
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.4 25 198.51.100.4 25
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.0.4 110 198.51.100.4 110
msk-donskaya-babdullakhi-gw-1(config)#ip nat inside source static tcp 10.128.6.200 3389 198.51.100.10 3389
msk-donskaya-babdullakhi-gw-1(config)#exit
msk-donskaya-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-babdullakhi-gw-1#
```

Рис. 6: Статический NAT и сохранение конфигурации

Проверка работоспособности

Проверка доступа из сети дисплейных классов

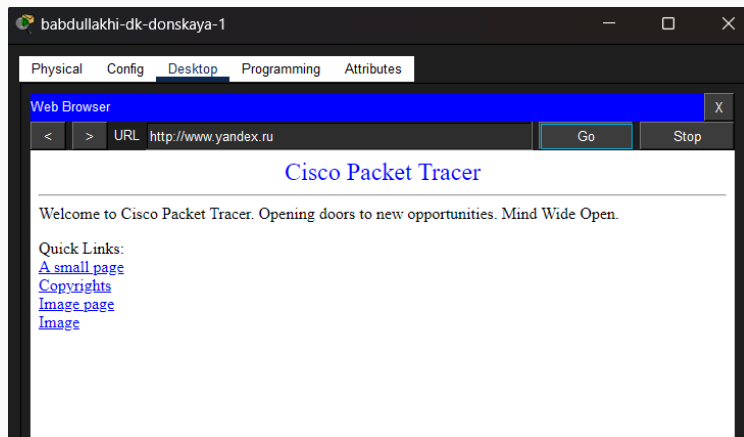


Рис. 7: Доступ к сайту www.yandex.ru

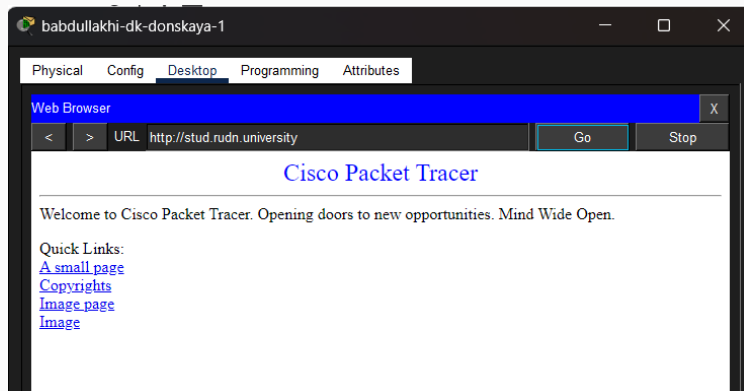


Рис. 8: Доступ к сайту stud.rudn.university

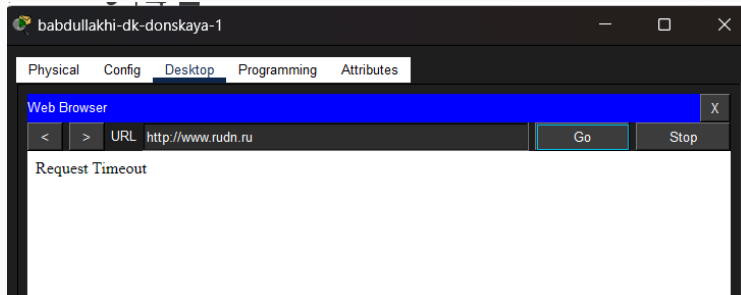


Рис. 9: Попытка доступа к недоступному ресурсу

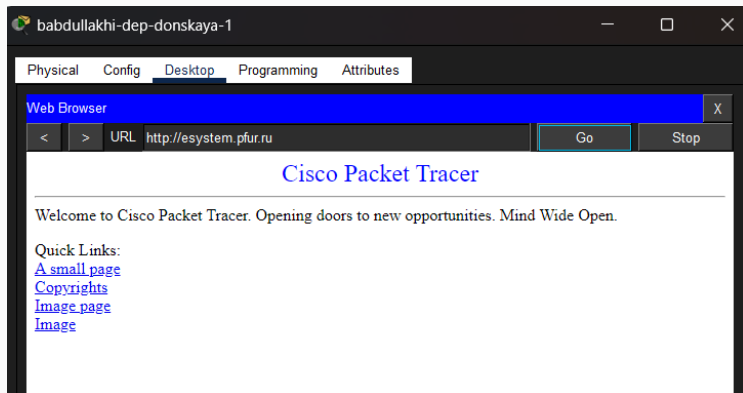


Рис. 10: Доступ к образовательному ресурсу

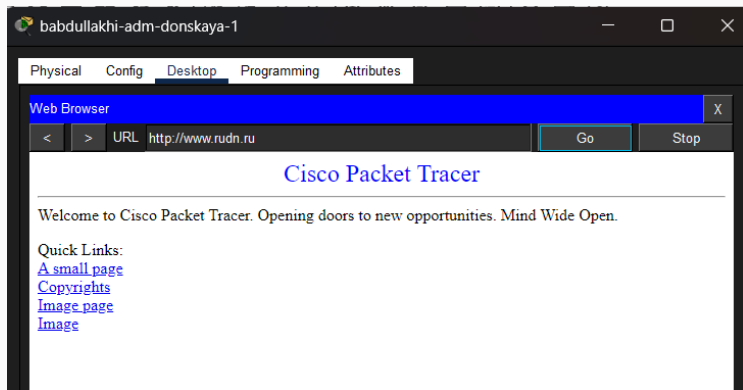


Рис. 11: Доступ к сайту университета

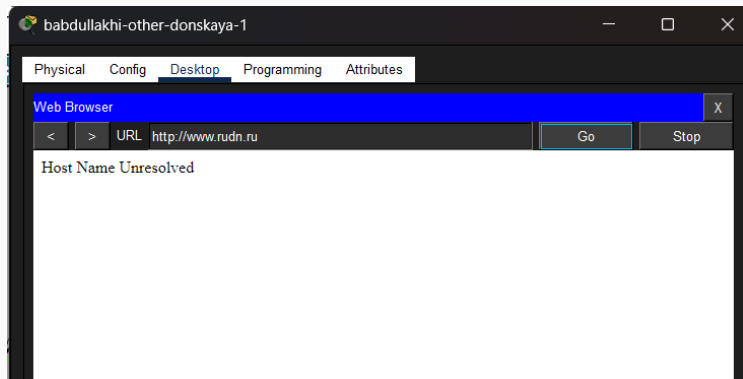


Рис. 12: Отсутствие доступа в Интернет у обычного пользователя

Проверка доступа администратора

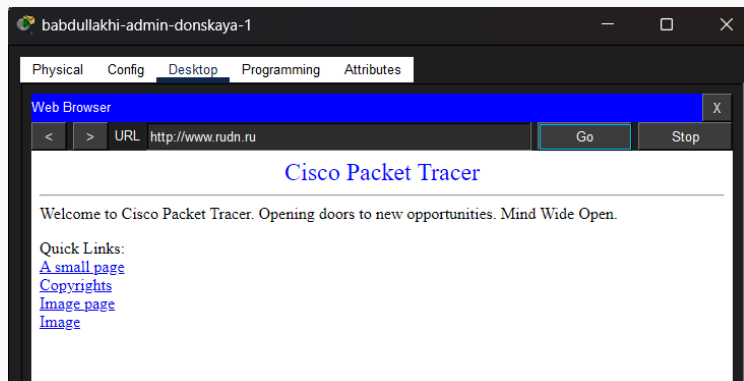


Рис. 13: Полный доступ администратора во внешнюю сеть

Проверка доступа к серверам

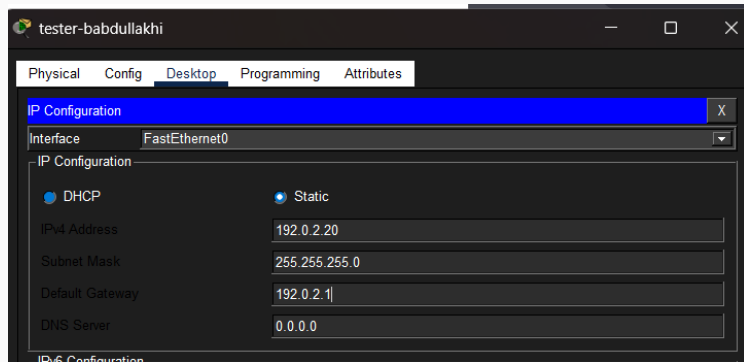


Рис. 14: Настройка IP-адреса тестового ПК

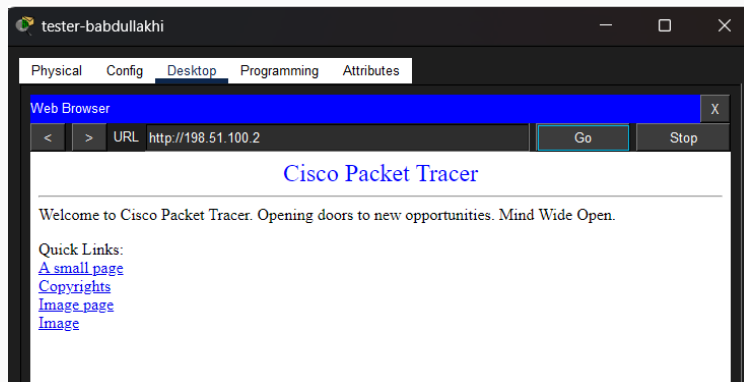
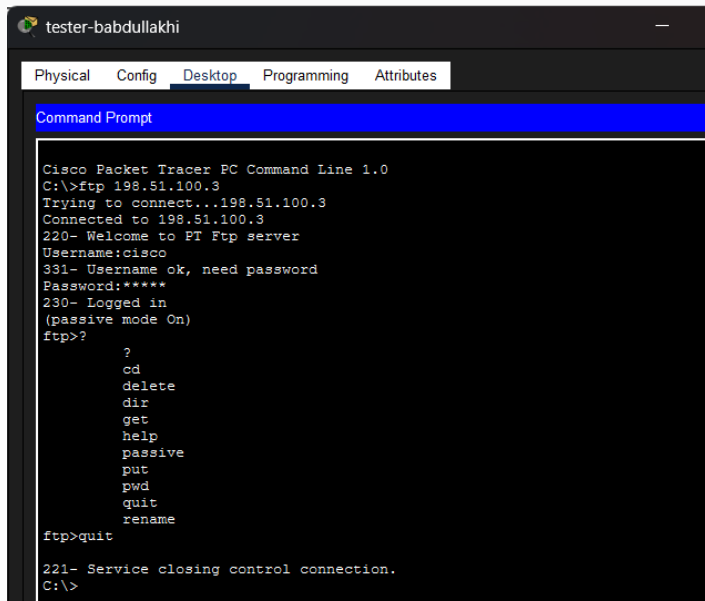


Рис. 15: Доступ к WEB-серверу по HTTP

Проверка FTP-сервера



The screenshot shows a Packet Tracer PC Command Line window for a device named 'tester-babdullakhi'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The 'Command Prompt' tab is active, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 198.51.100.3
Trying to connect...198.51.100.3
Connected to 198.51.100.3
220- Welcome to FT Ftp server
Username:cisco
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>?
?
cd
delete
dir
get
help
passive
put
pwd
quit
rename
ftp>quit
221- Service closing control connection.
C:\>
```

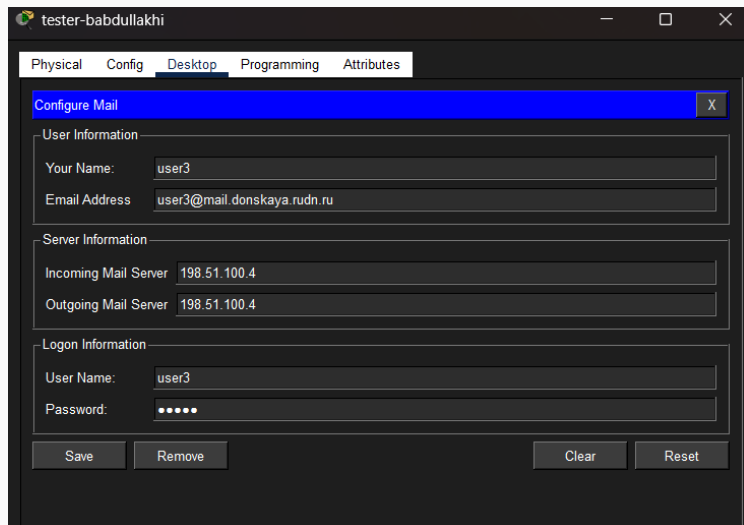


Рис. 17: Настройка почтового клиента

Проверка почтового сервера

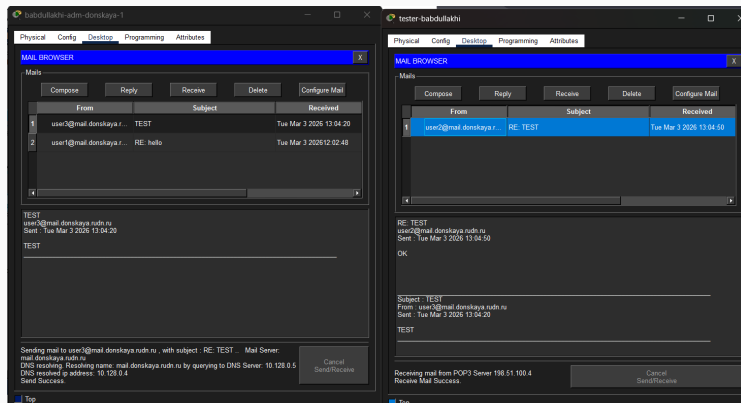


Рис. 18: Проверка отправки и получения почты

Выводы по проделанной работе

В ходе лабораторной работы была построена и настроена сеть с использованием технологии NAT в Cisco Packet Tracer. Были настроены маршрутизаторы и коммутаторы, подинтерфейсы VLAN, NAT-пул, списки доступа и статические преобразования NAT.

Проверка показала, что пользователи разных сегментов сети имеют доступ только к разрешённым ресурсам. Также была подтверждена корректная публикация внутренних серверов: WEB-сервер доступен по порту 80, FTP-сервер — по портам FTP, а почтовый сервер — по портам 25 и 110.